PROJECT TECHNICAL DOCUMENTATION

BioSpecInfo

Web platform for computational chemistry, physics and mathematics

Deploy Guide

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Deploy Guide — BioSpecInfo

Field	Value
Software	BioSpecInfo
Version described	`bsi-v250`
Purpose	Operational procedure for publishing, updating and rolling back the application.

0. What gets published, in practice

BioSpecInfo is a set of **static files**. There is nothing to compile, no process to keep running, no database to provision.

Build	None. The repository files *are* the application.
Server runtime	None
Environment variables	None for the application. Only the optional proxy has any (§4).
Database	None server-side
Server requirement	Serve static files over HTTPS with the correct MIME types

The one non-obvious requirement: the server must return `application/wasm` for `.wasm` files, otherwise RDKit and SQLite do not start. GitHub Pages, Netlify, Vercel and Cloudflare Pages already do.

1. Environments

Environment	How it is obtained	What it is for
Development	`python3 -m http.server 8899` from the root	Daily work and verification benches
Preview	A branch published on Pages, or `npx serve`	Checking before merging
Production	GitHub Pages from the `main` branch	`samupropio1-ship-it.github.io/BioSpecInfo-v11/`

> **Why a server is needed even in development.** Opening `index.html` with a double click (`file://`) does not work: RDKit and SQLite are WebAssembly modules and require an HTTP context. It is the reason the verification benches also run against a local server.

2. Publishing to production

2.1 Procedure

```
# 1. from the working branch, the full battery
python3 -m http.server 8899 &
node tools/genera-evidenza.js          # must close with 0 failures

# 2. bump the version — with the tool, NEVER by hand
node tools/porta-versione.js bsi-v<NNN+1> --prova   # look first
node tools/porta-versione.js bsi-v<NNN+1>           # then write

# 3. regenerate SBOM, evidence, PDFs and packages — IN THIS ORDER
#    The packages bundle the PDFs and the evidence: built first, they carry
#    the typeset documents and the report of the PREVIOUS version.
#    genera-pacchetti now notices and stops, but better not to get there.
node tools/genera-sbom.js
node tools/genera-evidenza.js       # the battery; needs a server on :8899
node tools/genera-pdf.js            # 36 PDFs; every footer carries version and commit
node tools/genera-pacchetti.js      # the three packages; fails on broken links, stale PDFs
or stale evidence

# 4. merge into main
git add -A && git commit -m "…"
git push -u origin <branch>
#    then pull request and squash merge
```

GitHub Pages publishes automatically from `main`: the *pages build and deployment* workflow takes about a minute.

2.1-bis Publishing the version's release

The application is online as soon as `main` updates; the **release** serves a different purpose: giving an assessor a fixed point to download — a tag, the version notes and the delivery packages attached.

It is published from the repository, not from outside:

> **Actions → release → Run workflow**, field `versione`: `bsi-v<NNN>`.

The workflow creates the tag on the current commit, opens the release and attaches the three packages from `consegna/`. The full archive — `models/` included, about 124 MB — is generated by GitHub itself as *Source code (zip)*.

Why from the repository and not from the command line. The credentials a remote session uses on the repository are authorised to write to branches and **not to tags**: the `POST` to `git-receive-pack` for `refs/tags/…` answers `403 Forbidden`, while a new branch goes through. The Actions token, by contrast, writes to the repository that hosts it.

What the workflow **refuses** to do:

Condition	Outcome
`CHANGELOG.md` has no `## [bsi-v<NNN>]` entry with content	fails: a release without notes looks like a version and does not say what changed
fewer than three packages in `consegna/`	fails: a missing attachment would make the release silently incomplete
the tag does not point at the commit the packages came from	fails: a release must have a single provenance

That last row exists because of a real defect. On re-running the workflow to refresh the packages, `gh release edit` updated notes and assets but **did not move the tag**: the *Source code (zip)* archive, which GitHub generates from the tag, stayed at the first publication's commit while the packages came from a later one. Two provenances, and nothing saying so. The tag is now moved, and a following step does not take that on trust: it asks GitHub where the tag points and compares.

The workflow is **re-runnable**: if the release already exists it updates notes, assets and tag instead of failing halfway. Moving an already-published tag is not free — whoever downloaded it already has different content — so it applies to a release just published, not an old one: for that, publish a new version.

2.1-ter Why the version bump is not done with a find-and-replace

It looks like the most harmless operation in the world: search `bsi-v<NNN>`, write `bsi-v<NNN+1>`. **For three releases in a row it was done that way, and every time it rewrote historical sentences** — sentences recording a fact that happened at a past version. What came out of it sat inside a signed conformance statement:

the document said	the truth
the password had been **changed at the current version**	it was changed at version `bsi-v188`
the tag had been **moved to the current version's commit**	the incident happened at version `bsi-v188`
the thirty-six entries were **added in the current version**	they were added in `bsi-v188`
"bump `CACHE` (e.g. `bsi-vN` → `bsi-vN`)"	an example equal to itself

No check could catch it: every line named a version that **exists**, in the document describing that version.

> This table uses placeholders rather than the real numbers because the two checks described below **would reject it** — which is the proof that they work.

`tools/porta-versione.js` touches **only** the lines that DECLARE the version — headers, footers, the README badge, package names, the two code lines — and **lists** everything else so it gets read. With `--prova` it writes nothing.

Two checks in `tools/verifica-documenti.js` now close the door: no past-tense sentence may name the current version, and no upgrade example may show the same version on both sides.

> **Name the thing, not the number.** "From the Pretsch tables" rather than "from version such-and-such": it reads better and does not age.

2.2 The two version lines

`CACHE` in `sw.js` and `BSI_APP_VERSION` in `index.html` **must match**.

- `CACHE` determines whether the Service Worker considers the cache stale:

without a bump, devices keep serving the old files and the publication reaches nobody.

- `BSI_APP_VERSION` is what the user sees in the "Updates" window. «Check for

updates» compares the active Service Worker's version (it answers the `BSI_VERSIONE` message with its `CACHE`) with the `var CACHE = '…'` line of the published `sw.js`, read from the same site with no cache and, only if that does not answer, from GitHub. `BSI_APP_VERSION` enters the comparison only when no Service Worker answers.

> **The shape of the line matters.** The check reads `sw.js` with the expression `var CACHE = '…'` . Up to version `bsi-v236` it looked for a `CACHE_VERSION` line that `sw.js` no longer contained, and the published version was never read. If the declaration in `sw.js` changes shape, `bsi-aggiorna.js` must change too; `test_controllaagg` notices.

They drifted apart once in the past — the second had been stuck for months — and the result was that the update check was comparing the wrong number.

2.2-bis Large files outside the precache

`models/spectra-noir.glb` (Spectra's face, about 8 MB) is not in the Service Worker's precache, like the other `models/*.glb`: it is downloaded the first time the app derives Spectra's icon or opens the Spectra Live console. It must be published with the rest, with its CC BY 4.0 attribution (`07-SBOM.md`); if it is missing, the console uses the fallback face.

2.2-ter One secret phrase, two fingerprints

Since version `bsi-v238` the same secret phrase serves the File Manager and the recognition of Spectra's creator. Its SHA-256 fingerprint is written in two places: `file_manager.html` and `bsi-spectra-creatore.js` (constant `IMPRONTA`). If the phrase changes, **both** must change in the same release; otherwise the File Manager opens with the new phrase and Spectra recognises only the old one. A recognition mark already written in a browser (`bsi-spectra-creatore-v1`) is not revoked by the change: it is removed with «forget that I am your creator» or with «Clear data». The plaintext phrase must never be written to a file in the repository.

2.2-quater A new module: three places to keep aligned

A new application `.js` file, such as `bsi-lezione.js` (the guided lesson), must be declared in `index.html` (with `defer`; `bsi-lezione.js` comes before `bsi-spectra-live.js`, which uses its `BSILezione` object when it receives «fammi lezione»), in the `PRECACHE` list of `sw.js`, so that it works offline, and in the module list of `tools/genera-sbom.js`, so that it appears in the SBOM. A module missing from the precache works while there is a network and disappears offline.

The latest module added this way is `bsi-spectra-esplora.js` (v242, «⚡ Explore with Spectra»): in `index.html` with `defer`, **after** `bsi-fisica-comune.js`, because it registers its three sections (`sesplchim`, `sesplfis`, `sesplmat`) with `BSIFisica.sezione` as soon as it runs; in the `PRECACHE` of `sw.js`, next to `bsi-spectra-spiega.js`; in the list of `tools/genera-sbom.js`. The three containers `<div id="sesplchim" class="section">`… and the three bar buttons are in `index.html`. The module brings no 3D files of its own: it uses the shared figure of `bsi-spectra-figura.js`.

In the same version `bsi-spectra-olointerattivo.js` (v242, the «⛶ Interagisci» interactive hologram) follows the same three rules: in `index.html` with `defer`, **right after** `bsi-spectra-figura.js`, which creates the button on the hologram and moves the figure into its window; in the `PRECACHE` of `sw.js`; in the list of `tools/genera-sbom.js`. It uses three.js and the figure already loaded by `bsi-spectra-figura.js` and the local assistant (`bsi-assistente.js`, `bsi-mat-cas.js`), and brings no data files of its own: molecules and planets are in the code.

2.2-quinquies Spectra Studio, a page of its own

`spectra-studio.html` (v239) does not go through `index.html`: it loads by itself `three.min.js`, `gltf_loader.js`, `bsi-spectra-mente.js`, `bsi-spectra-creatore.js`, `bsi-spectra-androide.js`, `bsi-spectra-corpo.js`, `bsi-spectra-rig.js`, `bsi-spectra-figura.js`, `bsi-spectra-voce.js` and `bsi-spectra-studio-lezioni.js`. The page and these modules are in the `PRECACHE` of `sw.js`; `bsi-spectra-figura.js` (v240) is also in `index.html`, with `defer`, because it shows Spectra with the body in the AI sections: it loads `three.min.js`, `gltf_loader.js` and the other 3D modules only the first time one of those sections is opened; `bsi-spectra-spiega.js` (v240, the «🧠 Spiega» tab of the Spectra window) is also in `index.html` with `defer` and in the precache; `bsi-spectra-voce.js` is also in `index.html`, because the console and the guided lesson use it. The studio's «Noir» face is the same `models/spectra-noir.glb` as the console's, outside the precache (§2.2-bis): if it does not arrive, the body stays without a face.

Spectra's body (v241): `models/spectra-corpo.bsi` must be deployed. The «Bounty Hunter» body (free on CGTrader, Royalty Free licence: used inside the product, not offered as a standalone file) is embedded in the app as the packed block `models/spectra-corpo.bsi` (4.40 MB), which is in the repository and must be deployed with the app: without it the figure and the studio stay on «Spectra non è arrivato: serve la rete la prima volta» with «↻ Riprova». The block is not a `.glb`: only `bsi-spectra-figura.js` (function `apri`) reopens it. Like the `models/*.glb` files it is not in the precache: the service worker (network first) caches it on the first request, and from then on it works offline. The original `.glb` (`models/spectra-corpo.glb`) stays out of git (`.gitignore`); the studio tries it only in development, if the block is missing. The `models/spectra-cyberpunk.glb` fallback no longer exists.

Regenerating the block (for example after changing the model):

```
node tools/impacchetta-corpo.js path/to/body.glb models/spectra-corpo.bsi
# ✓ models/spectra-corpo.bsi (4.40 MB), verificato: si riapre identico al .glb
```

The tool checks that the input is a `.glb`, writes the block («BSIC» · version · length · bytes mixed with an xorshift32 stream from the fixed seed, the same as in `bsi-spectra-figura.js`) and checks that it reopens identical. Then the new `models/spectra-corpo.bsi` is deployed; clients' caches refresh with the `sw.js` version change. A change of model must also be reflected in `models/CREDITS.md` and `THIRD_PARTY_NOTICES.md`. The quality voice calls `api.openai.com` directly from the browser: any `Content-Security-Policy` (`03-Security-Privacy-Compliance.md` §2.4) must allow it in `connect-src`, otherwise the voice falls back on the browser's synthesis.

2.3 How the update reaches users

1. The browser detects a different `sw.js` and installs the new Service Worker.
2. `skipWaiting()` activates it at once; the old caches are deleted.
3. The Service Worker **notifies** the open tabs (`BSI_SW_UPDATED`) — it does not

reload them on its own authority.

1. The page decides: if no work is in progress it reloads itself; if the

assistant is answering or there is unsaved text, it shows "Update now / Later".

Whoever opens the app at that moment already receives the new version.

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3. Publishing alternatives

The application does not depend on GitHub Pages. Any static file host will do, provided it is over HTTPS (the Service Worker requires it).

Platform	Command / configuration	Notes
GitHub Pages	Settings → Pages → branch `main`, folder `/`	In use. Free, custom domain possible
Netlify	`netlify deploy --prod --dir=.`	No build command
Vercel	`vercel --prod`	Static project, no framework
Cloudflare Pages	`npx wrangler pages deploy .`	Convenient if the proxy is already in use
VPS with nginx	Copy the files to `/var/www/biospecinfo`	See the configuration below

nginx configuration

```
server {
    listen 443 ssl http2;
    server_name biospecinfo.example.org;
    root /var/www/biospecinfo;

    # Without this MIME type, RDKit and SQLite do not start
    types { application/wasm wasm; }

    # sw.js must not be cached by the CDN: it is what announces new versions
    location = /sw.js {
        add_header Cache-Control "no-cache, no-store, must-revalidate";
    }

    location / {
        try_files $uri $uri/ /index.html;
    }
}
```

A rule that holds everywhere: `sw.js` must not be cached

If a CDN keeps `sw.js`, the browser keeps seeing the old version and **no update ever arrives**, however much one publishes. This holds on every platform.

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4. Optional proxy — a separate deployment

The proxy is a **distinct** component, owned by the user, published on their own Cloudflare account. It is not required in order to use the application.

```
cd proxy
npx wrangler login
npx wrangler deploy
# the address is printed: https://spectra-proxy.<account>.workers.dev

npx wrangler secret put GROQ_KEYS      # one or more keys, comma-separated
npx wrangler secret put GEMINI_KEYS    # optional
```

The address is then pasted into the assistant's "Proxy" box, inside the app. The same four steps are available **inside the application**, with the commands ready to copy.

Recommended configuration

```
# Without this variable the proxy accepts requests from any origin
npx wrangler secret put ORIGINI
# value: https://samupropio1-ship-it.github.io
```

Variable	Default	Why change it
`ORIGINI`	*(empty = all)*	**Always set it.** An open proxy is usable by anyone, at the expense of your quota.

Variable	Default	Why change it
`LIMITE_IP`	`20`	Requests per IP within the window
`TETTO_GIORNO`	`2000`	Overall daily ceiling

> **Keys must never go into `wrangler.toml`.** That file is versioned: a key written there ends up on GitHub, and the bots that scan it find it within hours. Use `wrangler secret put`, which keeps them out of the repository.

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5. Domain and certificate

Aspect	Situation
HTTPS	Mandatory: without it the Service Worker does not register and the app does not work offline
Certificate	Provided and renewed automatically by GitHub Pages
Custom domain	Settings → Pages → Custom domain, plus a `CNAME` record pointing to `<user>.github.io`

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6. Monitoring

There is no server to monitor: there are no processes that could fall over, nor resources that could run out.

What to watch	How
Outcome of the publication	GitHub → Actions → *pages build and deployment*
Availability	An HTTP check on `index.html` (UptimeRobot or similar)
Client-side errors	The browser console. Remote telemetry exists but is **disabled** (`BSI_TELEMETRY_URL` empty): enabling it requires a deliberate decision about data handling.
Proxy consumption	Cloudflare → Workers → Analytics

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7. Rollback and emergency procedures

7.1 Going back to the previous version

```
git revert <commit>
git push origin main
```

Pages republishes in about a minute. Devices receive the restored version as a normal update — **provided `CACHE` changes**: a revert that also takes the version number backwards is fine, because what counts is that the value be *different*, not that it be greater.

7.2 "I published but I do not see the changes"

In order:

1. **Was the version bumped?** It is the cause in ninety per cent of cases.

Without it, the Service Worker serves the cache.

1. **Has Pages finished?** GitHub → Actions.
2. **Is `sw.js` cached by a CDN?** It must answer `no-cache`.
3. From the app: ✨ menu → 🔄 **Updates → Check for updates**. It tells

which version is served and which is published; if they differ it forces a `registration.update()`. If it answers «up to date» with the old number, the `sw.js` the site serves is still the old one (points 2 and 3).

1. As a last resort: developer tools → Application → Service Workers →

Unregister, then reload.

7.3 "The assistant has stopped answering"

This is not a publishing problem: it concerns the AI provider. In the app, the 🔌 **Test** button measures in a few seconds which providers the device can actually reach, without needing any key.

7.4 Restoring the user's data

The data live on the device and **there are no copies on a server**: they cannot be restored remotely. This is a direct consequence of the local-first architecture, and it is declared to the user. Exports (notes, documents) remain the only form of backup.

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8. Pre-publication checklist

- ☐ `node tools/genera-evidenza.js` closes with **0 failures**
- ☐ `CACHE` in `sw.js` and `BSI_APP_VERSION` in `index.html` bumped and **equal**
- ☐ `node tools/genera-sbom.js` re-run
- ☐ `node tools/genera-pdf.js` re-run — otherwise the attachments describe the previous version
- ☐ `node tools/genera-pacchetti.js` re-run and closed with **0 broken links**
- ☐ Evidence report regenerated against the commit being published
- ☐ Any new deviations noted in `09-Release-Conformance-Statement.md` §4
- ☐ No conflict markers (`verifica_guida` checks this across 58 files)
- ☐ No API key in the versioned files

- ☐ The accessibility debt **has not grown** — if it has fallen, regenerate the

baseline with `node tools/verifica-accessibilita.js --aggiorna-riferimento` and record it in the commit

- ☐ Code coverage **has not fallen** — `audit_copertura` fails if it has

Document updated to version `bsi-v250`.